International Communications in Heat and Mass Transfer 158 (2024) 107859

Contents lists available at ScienceDirect

International Communications in Heat and Mass Transfer

journal homepage: www.elsevier.com/locate/ichmt

# Modeling on molten pool transport in laser deposition processes by Lattice Boltzmann method

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A R T I C L E I N F O

*Keywords:*
Lattice Boltzmann method
Molten pool transport
Marangoni flow
Ce addition
Laser-directed energy deposition

A B S T R A C T

It is crucial to understand the flow, heat and mass transfer occurring in the molten pool to predict and optimize of microstructure, morphology and defects in laser-directed energy deposition (L-DED). A comprehensive Lattice Boltzmann Method (LBM) has been proposed to model the complicated multiphase flow in the molten pool with a mass addition. In this method, a single-relaxation-time (SRT)-LB model with a mass source was adopted for the velocity field, and a two-relaxation-time (TRT)-LB model for the temperature field. The Partially Saturated Method (PSM) handled the non-slip condition on the solid-liquid interface. The Free Surface Lattice Boltzmann Method (FSLBM) captured the gas-liquid interface and addressed the Marangoni effect. The model has been validated through spot welding using high-speed camera, as well as single- and multi-pass deposition experiments. The effect of Ce addition into high-speed steel powders has been further investigated with the LBM model. Simulation results indicate that Ce addition changes the flow pattern from a single inward flow to a double-loop circulation flow. Simultaneously, the molten pool height decreases while the width increases. The double-loop circulation pattern increases the inclination angles between passes ($\omega$) and of the deposited layer ($\theta$), flattens the fusion line and the deposited layer surface.

## 1. Introduction

High-energy laser beams have been adopted in laser-directed energy deposition technology to melt metal powders layer by layer, enabling direct fabrication and repair of structural components. This technology finds extensive applications in aerospace [1], shipbuilding [2], and medical devices [3]. The molten pool serves as the pivotal issue for the melting and solidification within laser deposition processes. It significantly impacts the morphology [4], thermal history [5], and microstructure of deposited layers [6] with the characteristics of the limited spatial scale, high energy concentration, rapid phase changes, transient heat transfer, and dynamic behavior. In recent years, Computational Fluid Dynamics (CFD) methods, notably the finite volume method [7–9], have been extensively employed in the study of multiphase flows within the molten pool.

Compared to traditional CFD methods for simulating fluid flows, the lattice Boltzmann method exhibits significant advantages in handling complex boundary conditions in curved metal molten pools [10], and its inherent parallelizability substantially increases the computational efficiency of mesoscale models [11]. As a result, the lattice Boltzmann method has been widely adopted for mesoscale fluid dynamics models and extensively applied in additive manufacturing [12] and welding [13,14], providing in-depth studies on melting-solidification [15] and Marangoni flow [16]. Körner et al. [10] were the first to propose a lattice Boltzmann method capable of simulating the evolution of two-dimensional Powder Bed Fusion (PBF) molten pools, effectively addressing solid-liquid phase transitions, wetting, and surface tension. Building on this, Ammer et al. [17] extended Körner's model to three dimensions, constructing a comprehensive Electron Beam Melting (EBM) simulation model that includes a powder bed generation algorithm, an electron beam energy model, an energy absorption rate model, and a solid-liquid phase transition model. Subsequently, Rausch et al. [18] simulated multi-layer powder bed metal deposition, achieving up to 100-layer laser deposition.

However, the aforementioned models for molten pool evolution employ pre-laid powder bed models, which are not applicable to solving

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https://doi.org/10.1016/j.icheatmasstransfer.2024.107859

Available online 28 July 2024

molten pool evolution in laser-directed energy deposition processes involving powder feeding, as shown in Fig. 1. In existing lattice Boltzmann methods, two primary methods are used to manage fluid addition: one relies on specific boundary conditions [19,20], and the other involves introducing mass source or sink terms into the Navier-Stokes equations [21,22]. However, implementing fluid addition through new boundaries remains a challenge, particularly under complex boundary conditions. This approach might necessitate new interpolation or extrapolation schemes, thereby increasing model complexity and potentially affecting numerical stability. Therefore, for numerical simulations of laser-directed energy deposition processes, it is more appropriate to introduce mass source terms into the Navier-Stokes equations. In lattice Boltzmann methods, Halliday et al. [21] first introduced a method to incorporate mass source terms into the LB equations, utilizing a non-local scheme to calculate the spatial derivatives present in the source terms. Cheng et al. [23] developed an LB model that includes general source terms, also employing a non-local scheme for both temporal and spatial derivatives of these terms. Subsequently, Aursjø et al. [22] proposed an improved LB model with source terms that exclude temporal and spatial derivatives while maintaining Galilean invariance. However, these lattice Boltzmann algorithms with mass source terms are still in the early stages of research. Despite some research achievements in simulating mass generation [24] and gas-solid mass flow [25], there remains a lack of further development in coupling with other physical processes such as phase changes and Marangoni effect. Moreover, existing models typically treat the temperature field as a passive scalar within the flow [25], lacking integration with a mass source term model that accounts for the coupling between temperature and velocity fields. Regarding numerical applications, these models incorporating mass source terms have not yet been employed in additive manufacturing processes and require further investigation.

Therefore, this study innovatively proposes a coupled double-distribution-function lattice Boltzmann model with a mass source term. The model effectively addresses key physical processes in the additive manufacturing process, such as the addition of energy source terms, free surface capturing, Marangoni effect management, and prediction of heat transfer and flows within the molten pool. By incorporating mass source terms into the mass conservation equation, this model resolves the unique powder transport issues encountered in laser-directed energy deposition processes. Building on this model, the study further explores changes in the molten pool flow patterns induced by adding the rare earth element cerium (Ce) to T15 high-speed steel powder, comparing these with results from high-speed camera observations. Additionally, it investigates the impact of these flow patterns on the evolution of the molten pool velocity field and the macroscopic morphology of the deposited layers.

## 2. Numerical model

Based on the physical processes depicted in Fig. 1 and to ensure computational accuracy while reducing model complexity, the following assumptions were made for the numerical model: 1) Given the low dilution rate of the deposited layer observed in experiments, it is assumed that the substrate and deposited powder materials are identical; 2) The fluid in the metallic molten pool is considered to be an incompressible Newtonian laminar flow; 3) Considering the laser pulse separation time is significantly shorter than the computational time step, the laser is modeled as a continuous Gaussian heat source, disregarding convective heat transfer and thermal radiation between the molten pool and the environment; 4) Powder addition follows a continuous Gaussian distribution; 5) Recoil forces due to metal vaporization are neglected in the driving forces.

### 2.1. Governing equations

Based on the assumptions outlined above, the mass and momentum conservation equations are

$$\frac{\partial \rho}{\partial t} + \nabla \bullet (\rho \boldsymbol{u}) = S_m \tag{1}$$

$$\frac{\partial}{\partial t}(\rho \boldsymbol{u}) + \nabla \bullet (\rho \boldsymbol{uu}) = -\nabla p + \nabla \bullet [\mu(\nabla \boldsymbol{u} + \boldsymbol{u}\nabla)] + \boldsymbol{G} + \boldsymbol{F_b} + \boldsymbol{F_s} \tag{2}$$

Where $\rho$, $\boldsymbol{u}$, and $p$ represent density, velocity, and pressure, respectively; $S_m$ denotes the mass source term; $\mu$ is the dynamic viscosity; $\boldsymbol{F_s}$ is the surface tension, which varies with temperature; $\boldsymbol{G}$ represents gravity and is expressed as

$$\boldsymbol{G} = \rho \boldsymbol{g} \epsilon \tag{3}$$

Where $\boldsymbol{g}$ represents gravitational acceleration; $\epsilon$ represents volume fraction, which is used to distinguish gas-liquid cells. $\epsilon = 1$ indicates liquid nodes, $\epsilon = 0$ indicates gas nodes, and $\epsilon$ ranges from 0 and 1 at the gas-liquid interface. $\boldsymbol{F_b}$ is the buoyancy force, considered under the Boussinesq approximation, and given as

$$\boldsymbol{F_b} = -\rho \boldsymbol{g} \beta f_l (T - T_{ref}) \tag{4}$$

Where $\beta$ is the thermal expansion coefficient; $f_l$ denotes the liquid fraction, where $f_l = 1$ represents the liquid phase region and $f_l = 0$ represents the solid phase region; $T$ and $T_{ref}$ respectively indicate the local temperature and the reference temperature.

The energy conservation equation can be described as

$$\frac{\partial(\rho C_p T)}{\partial t} + \nabla \bullet (\rho C_p T \boldsymbol{u}) = \nabla \bullet (\lambda \nabla T) + q + S_h \tag{5}$$

Where $C_p$ represents the specific heat capacity at constant pressure, $\lambda$ denotes the thermal conductivity within the system, $S_h$ represents the

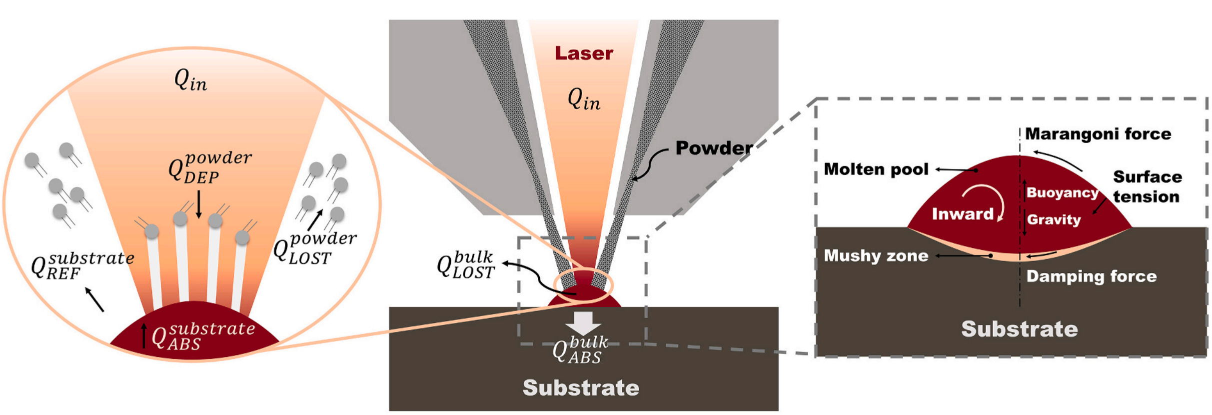



**Fig. 1.** Laser-directed energy deposition process and driving forces.

energy source term, and $q$ denotes the latent heat source term.

$$q = -\frac{\partial(\rho L f_l)}{\partial t} \tag{6}$$

Referencing Huang et al. [26] for the handling of the latent heat source term, the latent heat source term $q = -\partial(\rho L f_l)/\partial t$ is transposed to the left and combined with the time-varying term $\partial(\rho C_p T)/\partial t$. Eq.(5) is revised to Eq.(7) using Eq.(6):

$$\frac{\partial(\rho H)}{\partial t} + \nabla \bullet (\rho C_p T \boldsymbol{u}) = \nabla \bullet (\lambda \nabla T) + S_h \tag{7}$$

The total enthalpy $H$ is defined as $H = C_p T + L f_l$, where $C_p T$ represents the sensible enthalpy, and $L f_l$ denotes the latent enthalpy.

### 2.2. SRT-LB model for the incompressible flow field

To facilitate the addition of material during the laser cladding process and referencing Aursjø et al. [22] for the discretization of the mass source term, the SRT-LB method incorporating a mass source term is employed to solve Eq.(1) and Eq.(2). The evolution equation for the density distribution function $f_i$ is [22]

$$f_i\left(x + \boldsymbol{e}_i\delta_t, t + \delta_t\right) - f_i\left(x, t\right) = -\frac{1}{\tau_f}\left[f_i(x,t) - f_i^{eq}(x,t)\right] + \Delta\Omega_i^{F} + \Delta\Omega_i^{S} \tag{8}$$

Where $e_i$ denotes the discrete velocity in the direction $i$, $\tau_f$ is the dimensionless relaxation time. For three-dimensional incompressible flow, the D3Q19 model is adopted, defining the discrete velocities $\boldsymbol{e}_i$ and lattice weights $\omega_i$ by

$$\boldsymbol{e}_i = \begin{cases} (0,0,0) \\ c(\pm1,0,0), c(0,\pm1,0), c(0,0,\pm1) \\ c(\pm1,\pm1,0), c(\pm1,0,\pm1), c(0,\pm1,\pm1) \end{cases} \quad \omega_i = \begin{cases} 1/3 & i=0 \\ 1/18 & i=1,\dots,6 \\ 1/36 & i=7,\dots,18 \end{cases} \tag{9}$$

The discrete external force term is denoted by $\Delta\Omega_i^{F}$, the discrete mass source term is indicated by $\Delta\Omega_i^{S}$.The equilibrium distribution function is expressed in the following form, with $c_s = c/\sqrt{3}$. as a constant, representing the sound speed.

$$g_i\left(x + \boldsymbol{e}_i\delta_t, t + \delta_t\right) - g_i\left(x, t\right) = -\frac{1}{\tau_g^+}\left[g_i^+(x,t) - g_i^{eq,+}(x,t)\right] - \frac{1}{\tau_g^-}\left[g_i^-(x,t) - g_i^{eq,-}(x,t)\right] + \delta_t S_{h,i} \tag{18}$$

The discrete body force term $\Delta\Omega_i^{F}$ can be described as [27].

$$\Delta\Omega_i^{F} = \delta_t F_i = \omega_i \delta_t \left(1 - \frac{1}{2\tau_f}\right)\left[\frac{\boldsymbol{e}_i\cdot(\boldsymbol{G} + \boldsymbol{F_b} + \boldsymbol{F_s})}{c_s^2} + \frac{\boldsymbol{u}(\boldsymbol{G} + \boldsymbol{F_b} + \boldsymbol{F_s}) : (\boldsymbol{e}_i\boldsymbol{e}_i - c_s^2\boldsymbol{I})}{c_s^4}\right] \tag{10}$$

The discrete mass source term $\Delta\Omega_i^{S}$ is given by [22].

$$\Delta\Omega_i^{S} = \omega_i \delta_t \left(1 - \frac{1}{2\tau_f}\right)\left[1 + \frac{\boldsymbol{e}_i}{c_s^2} + \frac{(\boldsymbol{e}_i\cdot\boldsymbol{u})^2}{2c_s^4} - \frac{\boldsymbol{u}^2}{2c_s^2}\right] S_m \tag{11}$$

Where the mass source term $S_m$ is modeled using a Gaussian distribution, expressed as

$$S_m = \frac{\lambda_M \dot{M}}{\pi R_M^2 h}\left(exp\left(-\frac{\lambda_M r^2}{R_M^2}\right)\right) \Big/ \left((1 - exp(-\lambda_M))\right) \tag{12}$$

Where $\lambda_M$ represents the mass distribution coefficient, $R_M$ denotes the mass distribution radius, $r$ is the distance from the lattice unit to the center of the laser beam, and $h$ is the height of the lattice. The term $(1 - exp(-\lambda_M))$ acts as a normalization factor to ensure that the total mass of the distribution matches the actual added mass of the powder. $\dot{M}$ represents the rate of powder mass addition and is expressed as

$$\dot{M} = \rho A_{cross} V_{scan} \tag{13}$$

Where $A_{cross}$ represents the area of the deposition layer, and $V_{scan}$ represents the scanning speed of the laser beam.

The macroscopic quantities are calculated as follows

$$\rho\left(x,t\right) = \sum f_i + \frac{\delta_t}{2} S_m\left(x,t\right) \tag{14}$$

$$\rho\left(x,t\right)\boldsymbol{u} = \sum f_i \boldsymbol{e}_i + \frac{\delta_t}{2}\left((\boldsymbol{G} + \boldsymbol{F_b} + \boldsymbol{F_s}) + S_m(x,t)\boldsymbol{u}\right) \tag{15}$$

Given that the three force terms in Eq.(16) are independent of the velocity $\boldsymbol{u}$, the expression for velocity $\boldsymbol{u}$ is as follows:

$$\boldsymbol{u}\left(x,t\right) = \frac{1}{\sum f_i}\left[\sum f_i \boldsymbol{e}_i + \frac{\delta_t}{2}(\boldsymbol{G} + \boldsymbol{F_b} + \boldsymbol{F_s})\right] \tag{16}$$

By Chapman–Enskog analysis, the physical parameters in terms of the kinematic viscosity $\nu$ is given by [22].

$$\nu = c_s^{\,2}(\tau_f - 0.5)\delta_t \tag{17}$$

### 2.3. Enthalpy-based TRT-LB model for the temperature field

In laser deposition, a gas-liquid interface forms between the molten pool and the air. The laser beam energy cannot be directly incorporated as a boundary condition; instead, it is modeled as an energy source term ($S_h$) in the interface lattice units. The evolution equation of enthalpy distribution function $g_i$ is

Where, $\tau_g^+$ and $\tau_g^-$ are the dimensionless symmetric and antisymmetric relaxation times, respectively; $g_i^+(x,t)$ and $g_i^-(x,t)$ are the symmetric and antisymmetric distribution functions on direction $i$, respectively. The D3Q19 model is adopted for the temperature, in which $\boldsymbol{e}_i$ and $\omega_i$ are defined by Eq.(9). $g_i^{eq,+}(x,t)$ and $g_i^{eq,-}(x,t)$ are the corresponding equilibrium state distribution functions, expressed as

$$g_i^+(x,t) = \frac{g_i(x,t) + g_{-i}(x,t)}{2}, g_i^-(x,t) = \frac{g_i(x,t) - g_{-i}(x,t)}{2}$$

$$g_i^{eq,+}\left(x,t\right)=\frac{g_i^{eq}(x,t)+g_{-i}^{eq}(x,t)}{2},g_i^{eq,-}\left(x,t\right)=\frac{g_i^{eq}(x,t)-g_{-i}^{eq}(x,t)}{2} \tag{19}$$

Where $-i$ represents the opposite direction of $i$. For $i=0$, $g_0^+(x,t)=g_0(x,t)$ while $g_0^-(x,t)=0$, the equilibrium distribution function $g_0^{eq,+}(x,t)=g_0^{eq}(x,t)$ while $g_i^{eq,-}(x,t)=0$. The equilibrium function $g_i^{eq}(x,t)$ is

$$g_i^{eq}\left(x,t\right)=\begin{cases} H-(1-\omega_i)C_{p,ref}T, i=0 \\ \omega_i C_p T\left[\frac{C_{p,ref}}{C_p}+\frac{\boldsymbol{e}_i\cdot\boldsymbol{u}}{c_s^2}+\frac{(\boldsymbol{e}_i\cdot\boldsymbol{u})^2}{2c_s^4}-\frac{\boldsymbol{u}^2}{2c_s^2}\right], i\neq 0\end{cases} \tag{20}$$

Where $C_{p,ref}$ represents the reference specific heat capacity and is set as the harmonic mean of $C_{p,s}$ and $C_{p,l}$, calculated as $C_{p,ref}=2C_{p,s}C_{p,l}/(C_{p,s}+C_{p,l})$.

The discrete representation of the energy source term is defined as [28].

$$S_{h,i}=\left(1-\frac{1}{2\tau_g^+}\right)\omega_i S_h \tag{21}$$

Where the energy source term $S_h$ is modeled using a Gaussian ellipsoidal heat source model [29–31], expressed as:

$$S_h=\frac{6\sqrt{3}\dot{E}}{\pi\sqrt{\pi}R_aR_bR_c}\left(exp\left(-\lambda r_0{}^2\right)\right) \tag{22}$$

$$r_0{}^2=\frac{(x-x_0)^2}{R_a{}^2}+\frac{(y-y_0)^2}{R_b{}^2}+\frac{(z-z_0)^2}{R_c{}^2} \tag{23}$$

Where $\lambda$ is the energy distribution factor, $(R_a,R_b,R_c)$ represent the three semi-axes of the ellipsoidal model, $r_0$ denotes the relative distance between the lattice unit and the center of the laser beam, and $\dot{E}$ is the energy source term, which can be described as

$$\dot{E}=\alpha_t P \tag{24}$$

The macroscopic variable enthalpy $H$ can be calculated as

$$H=\sum g_i+\frac{1}{2}S_h\left(x,t\right)\delta_t \tag{25}$$

The temperature $T$ and liquid phase fraction $f_l$ are uniquely determined by the total enthalpy $H$.

$$T=T(H)=\begin{cases} T_s-\frac{H_s-H}{C_{p,s}}, H\leq H_s \\ \frac{H_l-H}{H_l-H_s}T_s+\frac{H-H_s}{H_l-H_s}T_l, H_s<H<H_l \\ T_l+\frac{H-H_l}{C_{p,l}}, H\geq H_l\end{cases} \tag{26}$$

$$f_l=f_l(H)=\begin{cases} 0, H\leq H_s \\ \frac{H-H_s}{H_l-H_s}, H_s<H<H_l \\ 1, H\geq H_l\end{cases} \tag{27}$$

Where, $T_s$ and $T_l$ respectively represent the solidus and liquidus temperatures, $H_s=C_{p,s}T_s$ and $H_l=C_{p,l}T_l+L$ are the corresponding enthalpy values. The thermal conductivity $\lambda$ and specific heat capacity $C_p$ are given by

$$\lambda=(1-f_l)\lambda_s+f_l\lambda_l \tag{28}$$

$$C_p=(1-f_l)C_{p,s}+f_lC_{p,l} \tag{29}$$

Where $\lambda_s$ and $\lambda_l$ respectively represent the thermal conductivity of the solid phase and the liquid phase.

By Chapman-Enskog analysis, The dimensionless relaxation time $\tau_g^-$ is determined using [32].

$$\frac{\lambda}{\rho C_{p,ref}}=c_s{}^2\left(\tau_g^- -0.5\right)\delta_t \tag{30}$$

To eliminate numerical diffusion across the phase interface, the two dimensionless relaxation time relations must satisfy the following relationship [33].

$$\frac{1}{\tau_g^+}+\frac{1}{\tau_g^-}=2 \tag{31}$$

### 2.4. Interfacial treatments

(i) Solid-liquid interface

The partially saturated method is employed to revise the lattice Boltzmann model for the velocity field at the solid-liquid interface, thereby ensuring the implementation of a no-slip condition at this interface. Eq.(8) is revised as [34].

$$f_i(x+e_i\delta_t,t+\delta_t)-f_i(x,t)=(1-B)\Omega_i^{fluid}+B\Omega_i^{solid}+(1-B)\delta_tF_i \tag{32}$$

Where $B$ is a weight coefficient dependent on $f_l$ and $\tau_f$,

$$B=\frac{(1-f_l)(\tau_f-0.5)}{f_l+\tau_f-0.5} \tag{33}$$

In Eq.(33), $\Omega_i^{fluid}$ represents the standard BGK collision operator for fluid ($f_l=1$) nodes, and is given by

$$\Omega_i^{fluid}=-\frac{1}{\tau_f}\left[f_i(x,t)-f_i^{eq}(x,t)\right] \tag{34}$$

In Eq.(33), $\Omega_i^{solid}$ represents a new collision operator applicable to semi-solid ($0<f_l<1$) and fully solid ($f_l=0$) nodes, and is given by

$$\Omega_i^{solid}=\left[f_{-i}(x,t)-f_{-i}^{eq}(\rho,\boldsymbol{u})\right]-\left[f_i(x,t)-f_i^{eq}(\rho,\boldsymbol{u}_s)\right] \tag{35}$$

Where, $\boldsymbol{u}_s$ is the velocity of solid phase movement, and in this study, $\boldsymbol{u}_s=0$.

From the above equations, it is evident that when $f_l=1$, the region is fluid, and the weight coefficient $B=0$, reducing Eq.(33) to Eq.(8); when $f_l=0$, the region is solid, and the weight coefficient $B=1$, transforming Eq.(33) into:

$$f_i(x+\boldsymbol{e}_i\delta_t,t+\delta_t)=f_{-i}(x,t)-f_{-i}^{eq}(\rho,\boldsymbol{u})+f_i^{eq}(\rho,\boldsymbol{u}_s) \tag{36}$$

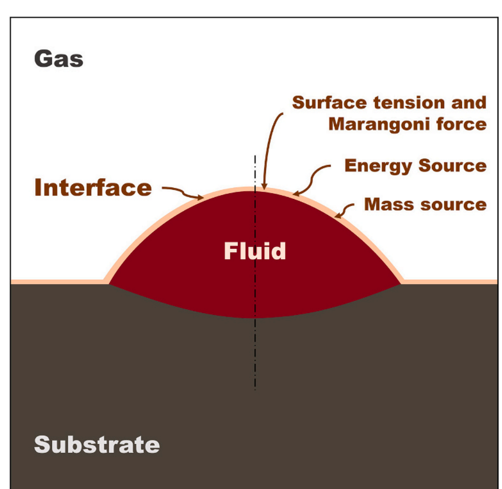



**Fig. 2.** Location of source terms and force terms.

The physical interpretation of this equation is to reflect the density distribution function $f_i(x,t)$ from solid nodes along the incident direction, which is equivalent to a $180^\circ$ transformation, thereby enforcing the no-slip condition ($\boldsymbol{u} = \boldsymbol{u}_s$) at the interface.

(ii) Gas-liquid interface

This study employs the free surface lattice Boltzmann method [35] to capture the gas-liquid interface, which is uniquely determined by the liquid volume fraction $\epsilon$. Here $\epsilon = 1$ indicates liquid nodes, $\epsilon = 0$ indicates gas nodes, and $\epsilon$ varies between 0 and 1 at the gas-liquid interface. Once the positions of the interface between the gas and liquid phases are determined, the surface tension model and corresponding source terms can be applied in the interface nodes, as depicted in Fig. 2.

Marangoni convection is considered the primary mechanism of mass transfer in a metal molten pool [36]. The surface tension model augmented with Marangoni force is expressed as

$$\boldsymbol{F}_s = -\left(\gamma\kappa\boldsymbol{n} - \nabla_s T\frac{\partial\gamma}{\partial T}\right)\delta_\Sigma \tag{37}$$

Where $\gamma$ represents the local surface tension parameter, $\boldsymbol{n}$ is the unit normal vector in the normal direction, $\nabla_s$ is the surface gradient operator, $\delta_\Sigma$ is the Dirac delta function, ensuring the continuous transition of surface tension at the gas-liquid interface, and $\kappa$ denotes the local surface curvature, calculated by

$$\nabla_s = (\boldsymbol{I} - \boldsymbol{n}\otimes\boldsymbol{n})\cdot\nabla \tag{38}$$

$$\kappa = -\nabla_s\cdot\boldsymbol{n} \tag{39}$$

The first term in Eq.(38) represents the surface tension of the fluid surface, acting in the normal direction, while the second term represents the Marangoni force generated by temperature gradients, acting in the tangential direction. The normal vector $\boldsymbol{n}$ and the Dirac delta function $\delta_\Sigma$ are determined by the gradient of the volume fraction $\epsilon$, calculated by

$$\boldsymbol{n} = \frac{\nabla\epsilon}{|\nabla\epsilon|} \tag{40}$$

$$\delta_\Sigma = \frac{1}{2}|\nabla\epsilon| \tag{41}$$

In thermal Marangoni flow, a state equation is required to describe the relationship between surface tension and temperature, which can be linear or nonlinear. Since the surface tension of the material system chosen in this study exhibits a linear relationship with temperature, this section will only discuss the impact of the linear relationship on model parameters.

$$\gamma(T) = \gamma_{ref} + \gamma_T\left(T - T_{ref}\right) \tag{42}$$

Where $\gamma_{ref}$ is the surface tension at $T_{ref}$, and $\gamma_T = \partial\gamma/\partial T$ denotes the rate of change of surface tension with respect to temperature. Substituting Eq.(39)-(43) into Eq.(38), the surface tension is obtained as

$$\boldsymbol{F}_s = -\left(\frac{1}{2}\gamma\kappa\nabla\epsilon - \frac{1}{2}\frac{\partial\gamma}{\partial T}|\nabla\epsilon|(\boldsymbol{I} - \boldsymbol{n}\otimes\boldsymbol{n})\cdot\nabla T\right) \tag{43}$$

The gradient terms of interface curvature κ, normal vector $\boldsymbol{n}$, and temperature gradient $\nabla T$ are all replaced by finite difference methods, as illustrated below [37].

**Table 1**
Compositions of two kinds of powders.

| | T15 | T15 + Ce |
|---|---|---|
| T15 (*wt*%) | 100 | 99.4 |
| $CeO_2$ (*wt*%) | 0 | 0.6 |

$$\frac{\partial\psi(\boldsymbol{x})}{\partial\boldsymbol{x}_a} = \frac{1}{c_s^2}\sum_i \omega_i\psi(\boldsymbol{x} + \boldsymbol{e}_i)e_{ia} \tag{44}$$

## 3. Materials and experiments

### 3.1. Powder materials

The experimental substrate is 42CrMo medium-carbon steel, while the deposited powder is a mixture of T15 high-speed steel and $CeO_2$. This mixed powder exhibits excellent wear resistance and, when used as a deposited layer on the workpiece, can significantly improve the workpiece's quality and lifespan [38]. Due to their significantly lower surface tension compared to metals, rare earth elements tend to segregate easily on the surface of the molten pool, thus reducing its surface tension. [39]. The addition of $CeO_2$ powder inevitably alters the surface tension of the molten pool, thus influencing the Marangoni flow and leading to changes in the morphology of the deposited layer. The compositions of the two types of deposited powders used in this study are listed in Table 1. The morphology of the powders is depicted in Fig. 3.

### 3.2. Laser deposition experiments

The laser deposition experiments utilize the TruDisk 4002 disk laser equipped with a Trumpf coaxial powder delivery laser head. A KUKA six-axis industrial robotic arm was used to precisely control the movement and positioning of the laser head along a predetermined trajectory. The laser metal deposition equipment is shown in Fig. 4.

This paper designed three types of experiments to verify the effectiveness of the numerical model: laser spot welding, single-pass, and multi-pass laser metal deposition. These experiments were conducted using two different powder models, T15 and T15 + Ce.

During the laser spot welding experiments, to avoid the interference of scattered deposition powder on the camera view, no powder feeding was conducted. The experimental substrate was an alloy block obtained by remelting alloy powder. The laser power for the spot welding experiments were set at 2000*W*, with an exposure time of 5 s.

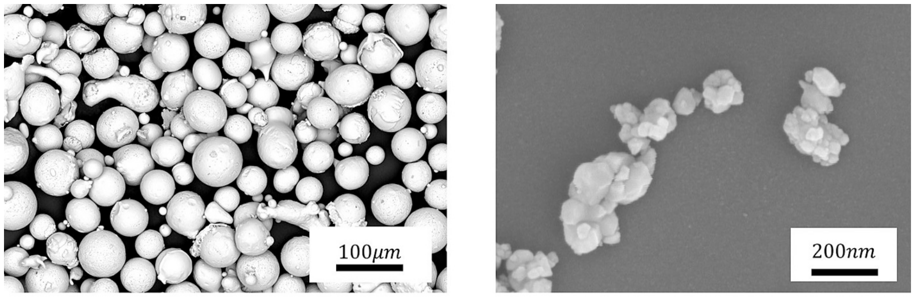
The flow of the molten pool during laser spot welding was captured using a high-speed camera system, shown as Fig. 5. The high-speed camera HX6 operated at a frame rate of 5000*fps*, and the captured results were processed using HX-Link software. Subsequently, the images captured were analyzed with VIC-2009 software to calculate the velocity field of the molten pool. VIC-2009 initially identified and recorded the positions of unmelted particles on the surface of the molten pool in different frames to determine the displacement of the captured area. It then combined the frame rate information to calculate the full flow velocity, which is characterized by high resolution and accuracy.

Both single-pass and multi-pass laser metal deposition experiments used 42CrMo steel as the substrate, measuring 10 × 10 × 10 mm. The substrates were prepared by wire cutting, followed by polishing with 200-grit sandpaper and cleaning with anhydrous ethanol. Before the deposition experiments, the deposition area was purged with pure argon gas for three minutes to eliminate oxygen from the environment. Argon gas was also used as the powder delivery medium during the experiments to transport metal powder to the substrate. The laser deposition process parameters are listed in Table 2. After the laser metal deposition experiments, a sample of the cross-section of the deposited layer was obtained by wire cutting. Following grinding, polishing, and etching, the macroscopic morphology of the deposited layer was observed using a Zeiss metallographic microscope.

## 4. Results and discussion

### 4.1. Code validation and grid independence

The current simulation has been conducted using a C++ − based




(a) T15 (b) $CeO_2$

**Fig. 3.** SEM images of T15 and CeO2 powders.

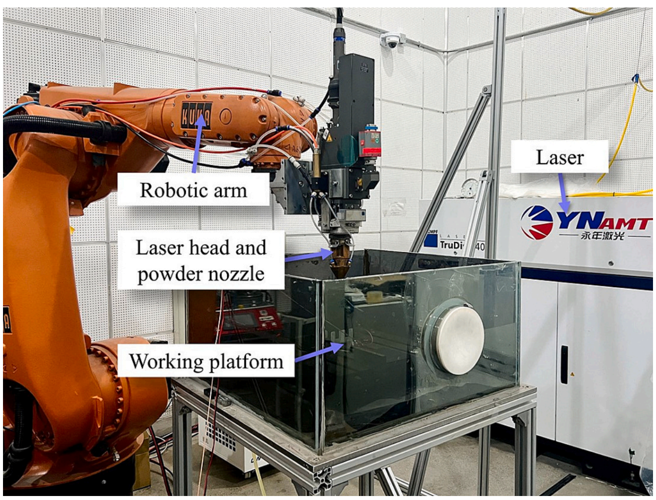



**Fig. 4.** Laser metal deposition devices.

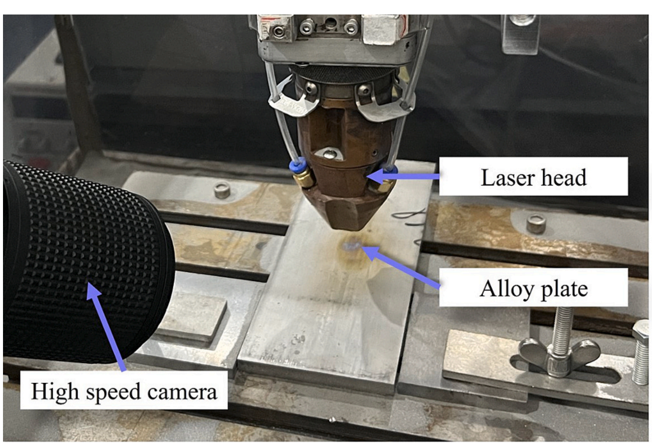



**Fig. 5.** High-speed camera system.

**Table 2**
The main experimental parameters of laser deposition.

| Parameter | Value |
|---|---|
| Laser power ($W$) | 2000 |
| Spot diameter ($mm$) | 4 |
| Argon gas velocity ($L/min$) | 16 |
| Feed rate ($g/min$) | 15 |
| Scanning speed ($mm/s$) | 7 |
| Overlap ratio (%) | 40 |
| The dwell time between passes ($s$) | 3 |

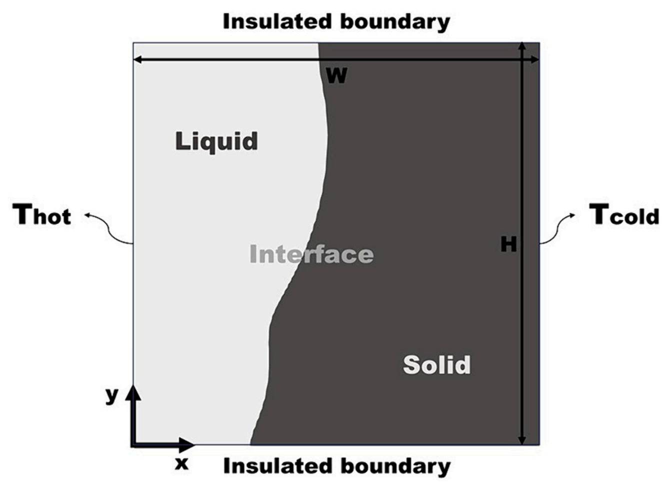



**Fig. 6.** Schematic of two-dimensional melting of pure substance.

lattice Boltzmann code. Due to the complexity of interface conditions and the challenges associated with incorporating source terms, relevant benchmark solutions have not been found in the literature. Given that the transport mechanisms within the molten pool during the DED process exhibit significant complexity, accurately simulating the phase change and flow process is fundamental for addressing other physical phenomena such as surface tension and powder addition. The classic two-dimensional cavity convection-driven melting problem was employed to validate the computational code and ensure grid independence. Fig. 6 gives the schematic of this two-dimensional melting problem by convection.

The simulation was conducted with $Pr = 0.021$ and $Ra = 2.25 \times 10^5$ for a cavity of $0.1m$ in height ($L$) with an aspect ratio ($W/H$) of 1. The cavity was filled with the metal tin, and its physical properties were consistent with those reported by Hannoun [40]. A grid independence analysis was performed with $75 \times 75$, $100 \times 100$, $125 \times 125$, $150 \times 150$, $175 \times 175$, and $200 \times 200$ lattice nodes. As shown in Fig. 7, The results were compared with those of Hannoun [40] at a grid size of $100 \times 100$. The comparison demonstrated a high degree of consistency regarding the vortex positions, sizes, and quantities at corresponding time steps.

The average Nusselt number on the heated left wall, $Nu_{ave}$, and the average liquid fraction within the cavity, $f_{l,ave}$, were calculated and are defined as follows:

$$Nu_{ave} = \frac{\int_0^H q_x dy}{\lambda(T_h - T_l)}\quad f_{l,ave} = \frac{\iint f_l dxdy}{W \cdot H} \tag{45}$$

Across varying grid resolutions, the average Nusselt number on the left heated wall and the average liquid fraction within the cavity exhibit trends consistent with Hannoun [40], as shown in Fig. 8. The observed small-scale, high-frequency oscillations in the Nusselt number may be

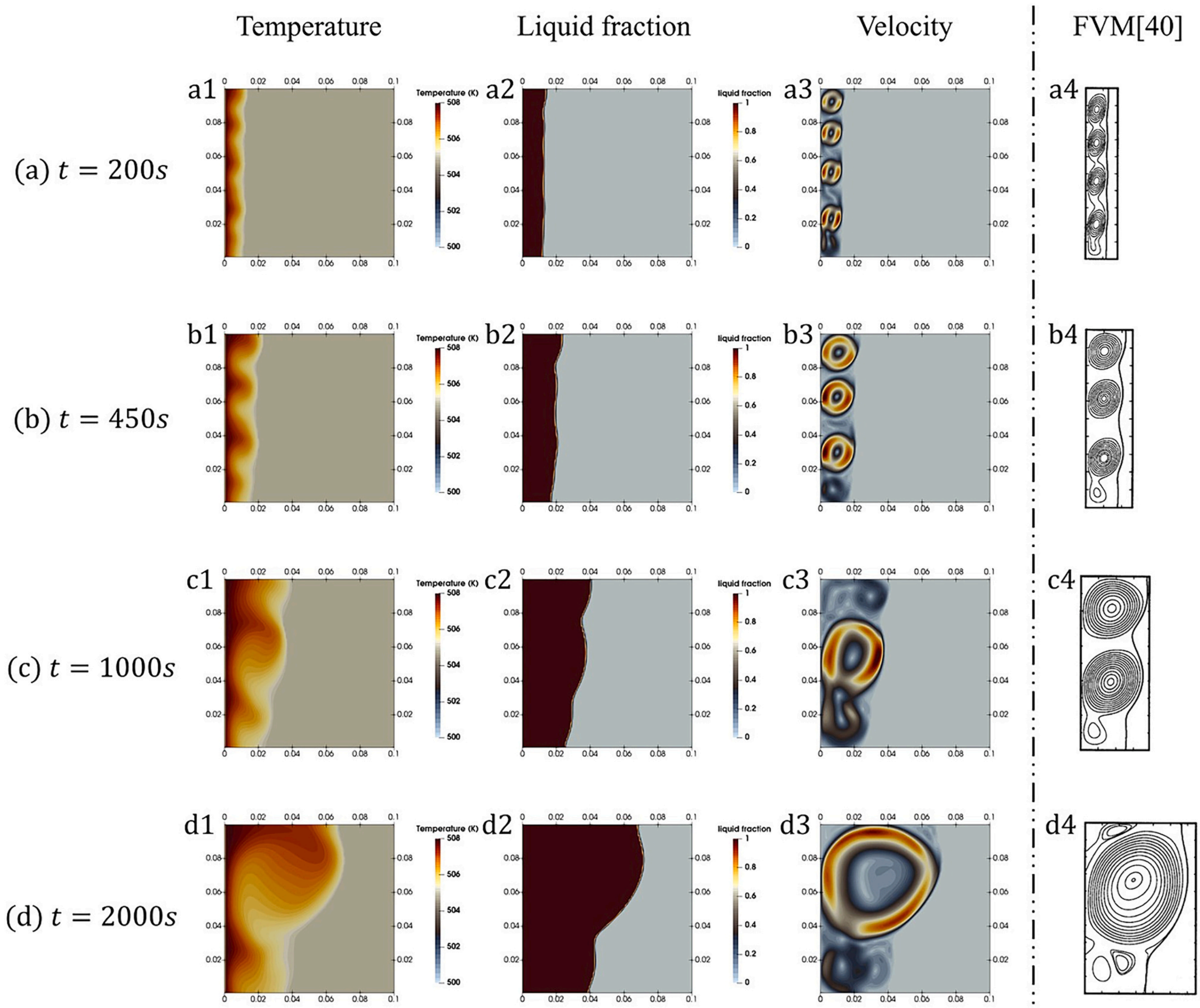



**Fig. 7.** Comparison of temperature, liquid fraction and velocity field with Hannoun [40].

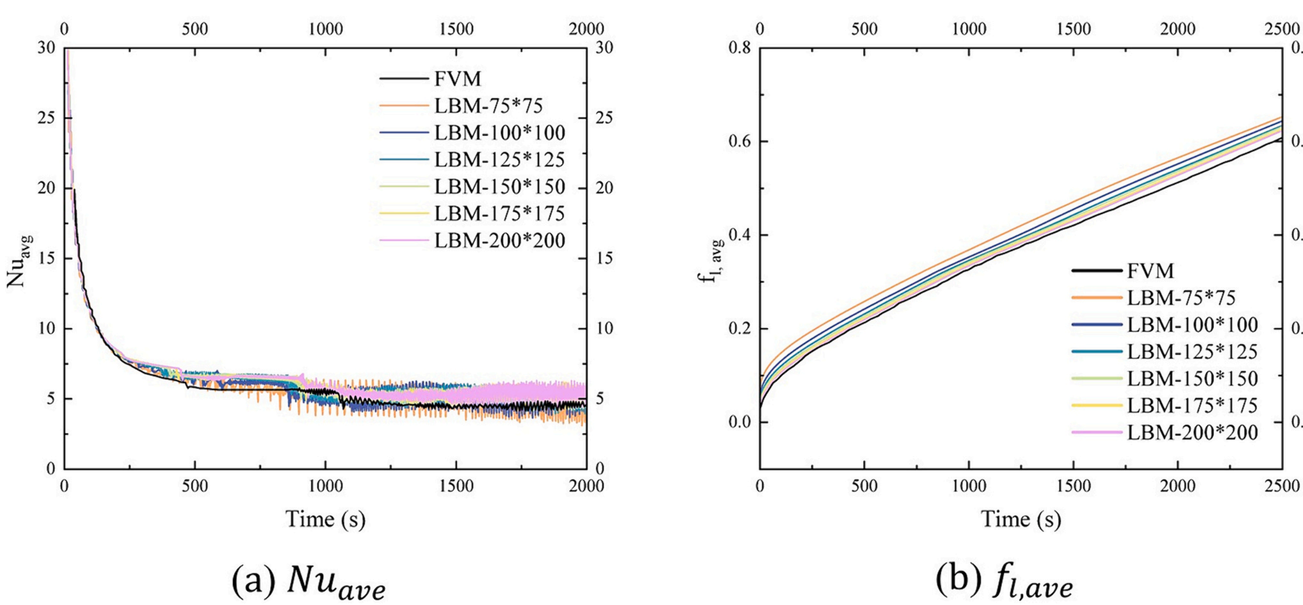



**Fig. 8.** Time evolution of the average Nusselt number and average liquid fraction.

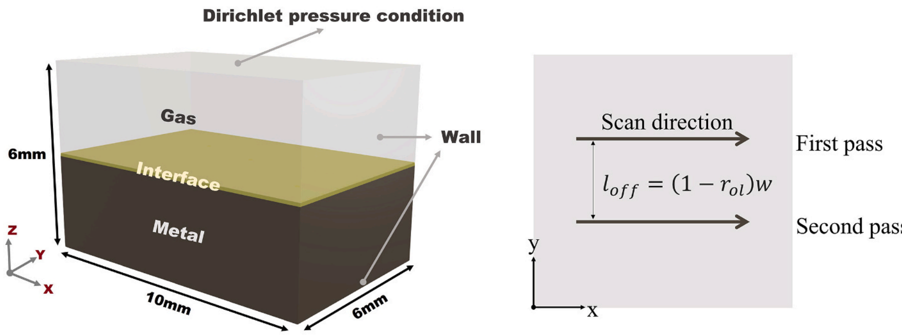



**Fig. 9.** Boundary conditions and multi-pass deposition strategy.

attributed to the instability of multi-vortex flow and the generation and disappearance of small vortices [26]. The maximum error in the average liquid fraction compared to Hannoun [40] was found to be 7.3%, indicating strong agreement. This confirms the accuracy of the model in simulating phase change processes.

### *4.2. Spot welding and laser deposition simulation model parameters*

Both the spot welding and laser deposition numerical models employed a uniform lattice with a lattice size of $\delta_x = 0.15mm$. In the upper surface of the computational domain, a Dirichlet pressure condition was set by $p = 0$ (i.e. $\rho = 1$), while the sides and bottom were set to Wall boundary conditions. The initial temperature condition is $T_0 = 300K$, as shown in Fig. 9(a).

No mass source term was added in the spot welding numerical model, and the energy source term was fixed at the geometric center of the interface. In the laser deposition model, the center of the energy and mass source terms moves over time. The parameter settings were consistent with those used in the laser deposition experiments. Fig. 9(b) depicts the scanning strategy for multi-pass deposition, with a deposition offset between passes denoted as $l_{off}$. This deposition offset can be calculated from the deposition width ($w$) and overlap ratio ($r_{ol}$), as derived from experimental data.

To validate the effectiveness and predictive accuracy of the established laser deposition model, the same two sets of powders used in the experiments were adopted. Both process parameters and material parameters were kept consistent with those used in the experimental model. The material parameters for the two types of deposition powders are presented in Table 3.

**Table 3**
Material properties.

| Properties | Model | Value |
|---|---|---|
| Solidus point $(K)$ | T15 | 1626 |
| | T15 + Ce | 1653 |
| Liquidus point $(K)$ | T15 | 1653 |
| | T15 + Ce | 1693 |
| Density $(kg/m^3)$ | | 7850 |
| Effective viscosity $(Pa \bullet s)$ | | 0.06 |
| Latent heat $(kJ)$ | | 250 |
| Effective thermal conductivity of solidus$(W/(m \bullet K))$ | | 40 |
| Effective thermal conductivity of liquid$(W/(m \bullet K))$ | | 209.2 |
| Surface tension $(mN/m)$ | T15 | $\gamma = 1.21T - 576\ (T > 1653K)$ |
| | T15 + Ce | $\gamma = 6.71T - 9715\ (1693K < T < 1789K)$<br>$\gamma = -0.23T + 2700\ (T > 1789K)$ |

### *4.3. Flow pattern in molten pool*

Fig. 10 displays the high-speed videography results of laser spot welding and the resulting videos can be found in Supplementary Video 1 and Video 2. Observations of the displacement of unmelted particles and bubbles on the molten pool surface reveal that, driven by surface tension, the liquid metal in the T15 molten pool flows toward the center, exhibiting a uniform inward flow tendency, and eventually converges at a bright central flow point. In contrast, the liquid alloy in the T15 + Ce molten pool demonstrates a double-loop circulation flow trend, with movements both outward from the center and inward from the edges.

The experimental velocity distributions along the x-axis were analyzed using VIC-2009 software to generate velocity contour maps, as shown in Fig. 11(a1) and Fig. 11(b1). Simultaneously, x-axis velocity contour maps of the gas-liquid interface from LBM simulation results were extracted, as depicted in Fig. 11(a2) and Fig. 11(b2). The spot welding simulation videos can be found in Supplementary Video 3 and Video 4. The flow patterns of T15 and T15 + Ce depicted in these contour maps are consistent with the qualitative observations in Fig. 10. A comparison with high-speed camera results indicates that the molten pools in the LBM simulations exhibit consistent left-right symmetry and stable flow, indicating good stability.

The flow pattern differences between these two types of molten pools can be attributed to the variations in surface tension models used for different deposition powders. In the T15 molten pool, the positive surface tension coefficient ($\partial\gamma/\partial T = 1.21$) results in the liquid metal flowing from cooler to hotter areas under the influence of the Marangoni force, leading to an inward flow tendency. In contrast, the T15 + Ce molten pool exhibits a transition temperature ($T_x = 1789K$), as shown in Table 3. At the edges of the molten pool, where the temperature is below $T_x$, a positive surface tension coefficient ($\partial\gamma/\partial T = 6.71$) promotes the inward flow of the liquid alloy. Conversely, in the central area where the temperature exceeds $T_x$, a negative surface tension coefficient ($\partial\gamma/\partial T = -0.23$) induces the liquid alloy to flow outward. This double flow behavior results in the distinct flow dynamics observed in the T15 + Ce molten pool.

To quantitatively compare the LBM simulation results with those from the high-speed camera experiments, horizontal flow velocities along the x-axis were extracted from Fig. 11(a1, a2) and Fig. 11(b1, b2), and the corresponding velocity curves were plotted in Fig. 11(a3, b3). In the T15 molten pool, within the region where $x < 0$, the horizontal flow velocity of the liquid metal initially increases, reaching a peak at point E1, and then gradually decreases to zero. This decrease occurs as flows from different directions converge near the center of the molten pool, collide, and slow down, resulting in a reduction in flow speed. The flow trend and velocity changes in the simulations closely mirror those observed experimentally. The experimental turning point E1 is at $x = -0.721mm$ with a maximum velocity $v_{E1,max} = 0.492m/s$; the simulation turning point S1 is at $x = -0.695mm$ with a maximum velocity $v_{S1,max} =$

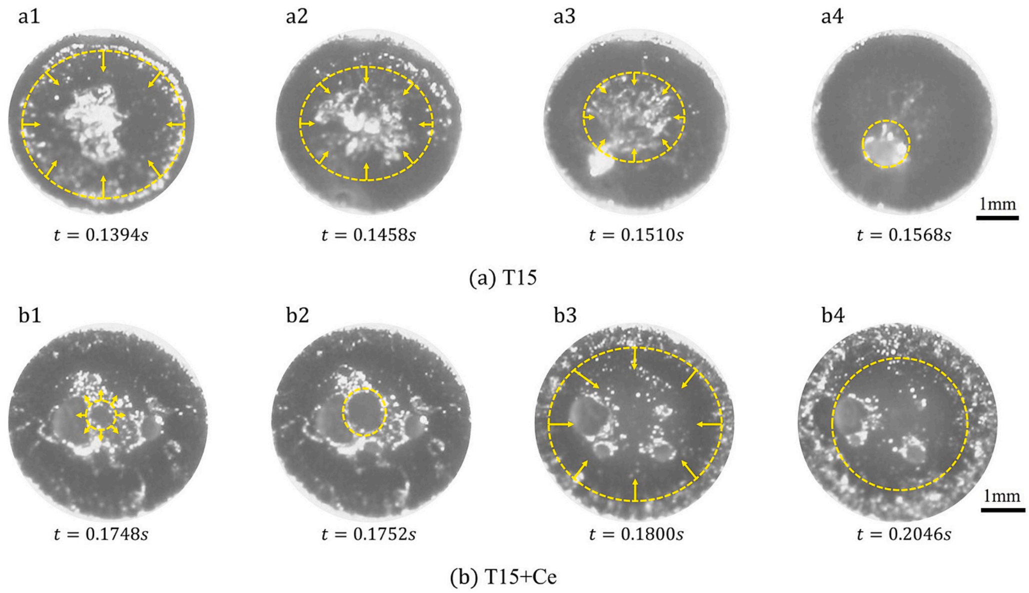



**Fig. 10.** The flow in the laser spot welding pool.

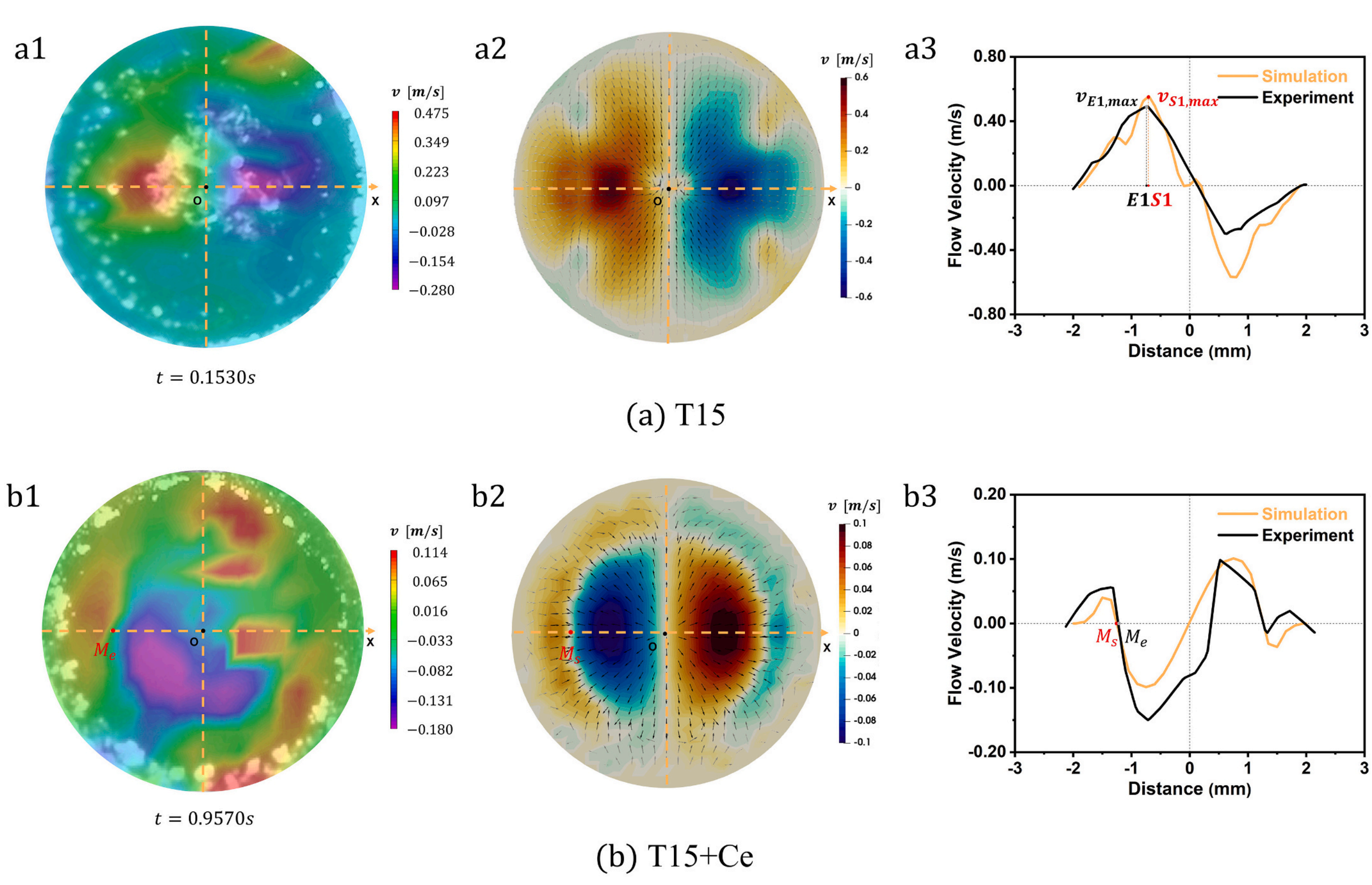



**Fig. 11.** Velocity distribution in molten pools.

$0.553m/s$, indicating good consistency.

In the T15 + Ce molten pool, the inward flow from the edges and the outward flow from the center converge at point $M_s$, where the flow becomes relatively stationary (with a horizontal velocity of zero). The convergence point $M_s$ from the simulation closely matches the experimental point $M_e$, indicating a similarity in the temperature field distributions between the simulation and experiment. Specifically, the proportions of high-temperature areas ($T > 1789K$) and low-temperature areas ($T < 1789K$) are consistent. This further validates the accuracy of the model.

Comparing the horizontal velocities of the T15 and T15 + Ce molten pools reveals that the maximum flow speed in the T15 molten pool ($v_{S1,max} = 0.553m/s$) is higher than that in the T15 + Ce molten pool ($v_{S2,max} = 0.101m/s$). This phenomenon is attributed to the double-loop

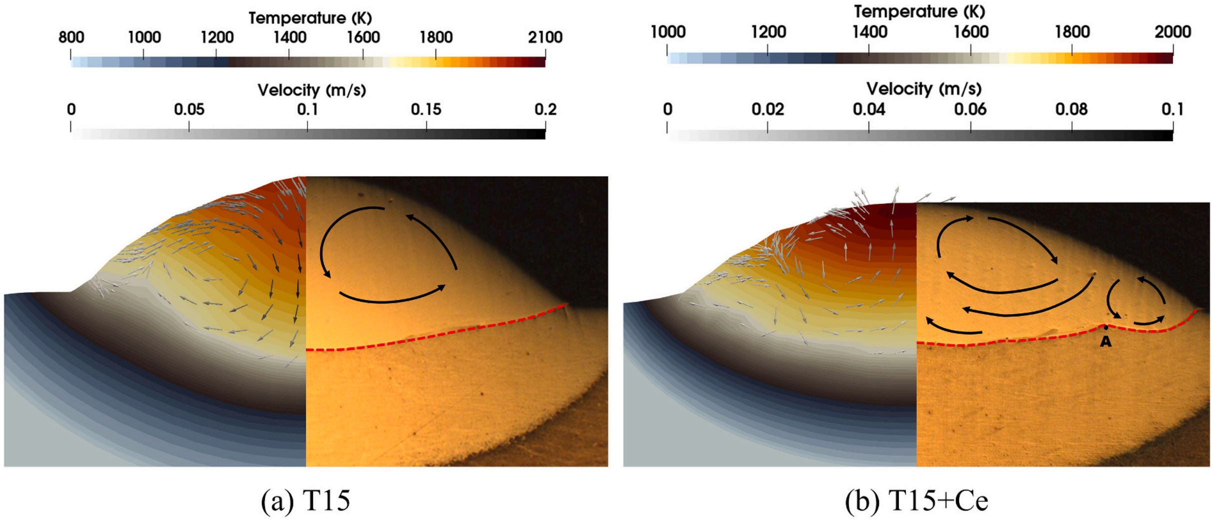



**Fig. 12.** Cross section distribution of flow and temperature in molten pool.

circulation pattern in the T15 + Ce molten pool. The collision between these inward and outward flows results in a reduction of the tangential velocity of the liquid metal, causing it to be pushed to the bottom of the molten pool before achieving sufficient acceleration. Consequently, the T15 + Ce molten pool exhibits a substantially lower velocity compared to the T15 molten pool.

### *4.4. Single-pass laser deposition*

(i) Molten pool flow pattern

Fig. 12 presents a comparative view of the cross-sections of T15 and T15 + Ce molten pools during a single-pass laser deposition process, comparing simulation results with experimental observations. Details of the Single-pass laser deposition simulation can be found in Supplementary Video 5 and Video 6. The flow patterns in molten pools are consistent with the results from spot welding experiments.

The relative magnitude of the inward and outward vortices in the T15 + Ce molten pool is determined by the temperature range below and above $T_x$. The simulation results (shown in Fig. 12(b)) indicate that the temperature range in the high-temperature area is slightly larger than that in the low-temperature area, resulting in slightly larger outward vortex currents than those of the inward ones. Experimental observations have also noted a similar phenomenon: at the bottom of the molten pool, a small protrusion formed at point A at the bottom of the molten pool due to the separation of inward and outward vortices. The location of point A in Fig. 12(b) indicates that the inward flow region of the molten pool in the experimental results is also smaller than the outward flow region, which further confirms consistency in flow characteristics.

(ii) Macroscopic morphology of deposition layers

To quantitatively describe the characteristics of the deposited layer morphology, the width $w$, height $h$, and height-to-width ratio $(h/w)_{Exp}$ of the deposition layer were extracted from the experimental data, as listed in Table 4. Additionally, the average height-to-width ratio $(h/w)_{Sim}$ during the stable deposition phase of the single-pass simulation was also extracted. A smaller $h/w$ ratio indicates a flatter deposition layer morphology.

**Table 4**
Key parameters of molten pool.

| | Average molten width | Average molten height | $(h/w)_{Exp}$ | $(h/w)_{Sim}$ |
|---|---|---|---|---|
| T15 | 3.70 mm | 0.98 mm | 0.265 | 0.252 |
| T15 + Ce | 4.03 mm | 0.76 mm | 0.188 | 0.198 |

Fig. 13 displays the half-sectional views of the molten pools for both T15 and T15 + Ce, cut along the central longitudinal section. According to LBM results, the inward flow in the T15 molten pool forms a narrow and high wedge-shaped molten pool, creating a relatively steep free surface on the deposited layer. During the stable deposition phase of T15 in the simulation, the average height-to-width ratio is 0.252, closely matching the experimental width-to-height ratio of 0.265 with an error of 5.0%. In contrast, the T15 + Ce molten pool forms a wider and lower molten pool, resulting in a flatter deposition layer. The average height-to-width ratio for T15 + Ce in the simulation is 0.198, closely matching the experimental width-to-height ratio of 0.188 with an error of 5.9%.

The flow pattern within the molten pool is a primary factor influencing changes in the morphology of the deposited layer. As illustrated in Fig. 13, in the T15 molten pool, inward flow concentrates the heat and powder from the edges to the center, resulting in a higher molten height and narrower molten width. In the T15 + Ce molten pool, the collision between inward and outward flows reduces the “climbing” capability of the molten pool at the edges, which results in a lower molten height. Concurrently, the formation of a high-temperature outflow area promotes the redistribution of heat within the molten pool, pushing most of the heat outward and significantly increasing the molten width $w$, ultimately creating a wide and flat molten pool. Additionally, in the T15 molten pool influenced by the inward flow, the high-temperature zone is deeper with a shorter longitudinal trailing. In contrast, in the T15 + Ce molten pool dominated by the outward flow, the morphology is flatter with a longer longitudinal trailing.

### *4.5. Multi-pass laser deposition*

Fig. 14 depicts the main geometric parameters of the multi-pass laser deposition process. Here, $\alpha$ represents the inclination angle of the molten pool, and $\beta$ represents the inclination angle of the flow vortex, both obtained from simulations. $\theta$ represents the inclination angle of the deposited layer, which is the angle between the deposited layer and the substrate, while $\omega$ represents the inclination angle between deposited passes. The parameters $h$ and $w$ respectively represent the deposition height and width, and these values are obtained from experimental results. The comparison between the simulated and experimental geometric parameters is presented in Table 5.

(i) Molten pool flow pattern

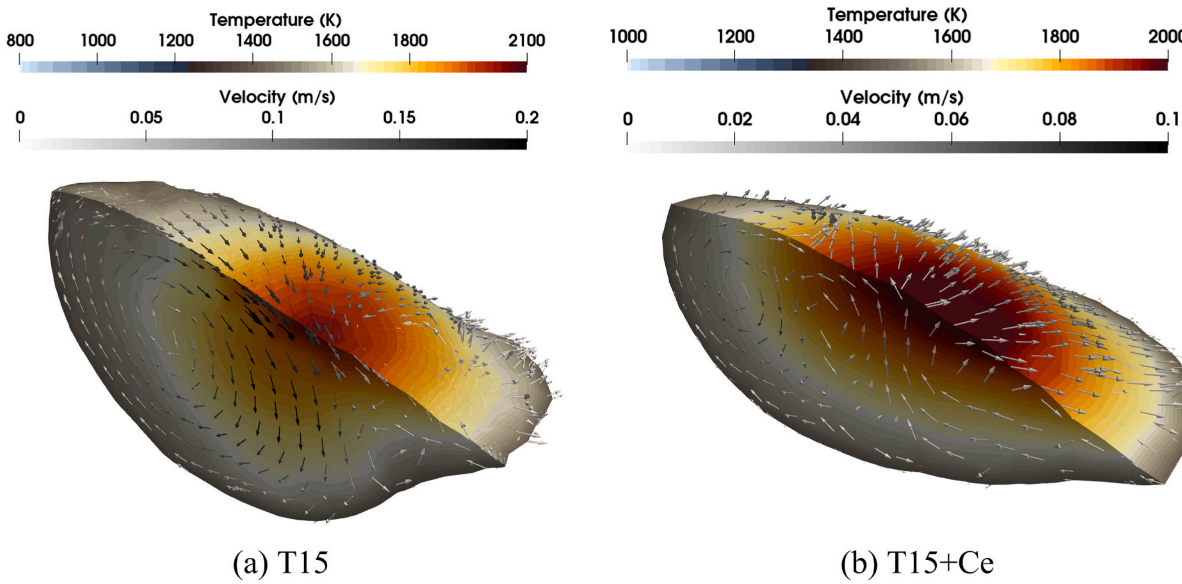



**Fig. 13.** Central longitudinal distribution of flow and temperature in molten pool.

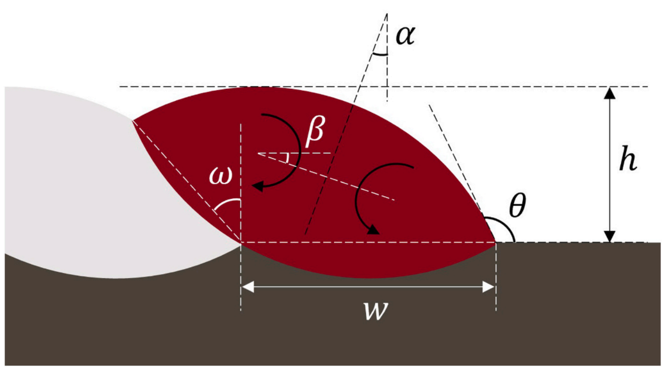



**Fig. 14.** Schematic representation of the geometric parameters of the multi-pass laser-deposited layer.

**Table 5**

Geometric parameters of multi-pass laser deposition.

| | Geometric parameters | T15 | T15 + Ce |
|---|---|---|---|
| LBM | Inclination angle of the molten pool ($\alpha$) | 9° | 21° |
| | Inclination angle of the flow vortex ($\beta$) | 16° | 20° |
| Experiment | Inclination angle between deposited passes ($\omega$) | 40° | 54° |
| | Inclination angle of deposited layer ($\theta$) | 140° | 151° |
| | Deposition width ($w$) | 3.81 *mm* | 4.29 *mm* |
| | Deposition height ($h$) | 1.52 *mm* | 1.17 *mm* |
| | $h/w$ | 0.399 | 0.273 |

Fig. 15 illustrates the temperature and velocity fields in the cross-sectional views of the multi-pass molten pool. To quantitatively describe the asymmetry between the T15 and T15 + Ce molten pool, the inclination angles $\alpha$ and the flow vortex inclination angles $\beta$ were extracted. For the T15 + Ce molten pool, $\alpha$ and $\beta$ are 21° and 20°, respectively, both of which are greater than those for the T15 molten pool, at 9° and 16°, respectively. This indicates that the asymmetry in the T15 + Ce molten pool is more pronounced.

The inclination angle $\alpha$ of the T15 molten pool results from the overlapping of successive passes, where accumulated metal impedes vortex development, leading to asymmetrical shapes. In the T15 + Ce molten pool, the high-temperature area on the left side (L1, yellow area) needs to climb, exhibiting a lower tangential speed. This permits the inward flow vortex at the cooler left edge (L2, yellow area) to fully develop. Consequently, the central high-temperature outflow area (L1, yellow area) becomes compressed, increasing the inclination angles $\alpha$ and $\beta$. Additionally, driven by both gravity and surface tension, the high-temperature area on the right side (R1, blue area) exhibits an extremely high outflow speed, which limits the development of inward flow on the right. This further increases the molten pool inclination angle $\alpha$.

(ii) Macroscopic morphology of deposition layers

Fig. 16 compares the morphology of the T15 and T15 + Ce molten pools with experimental results. Observations from Fig. 16(a2, b2) show that the T15 presents a narrow and steep deposition morphology, while T15 + Ce exhibits a wider and more gradual deposition morphology. The morphology in the multi-pass laser deposition simulation can be found in Supplementary Video 7 and Video 8. To quantitatively analyze the differences in deposition morphology between T15 and T15 + Ce, measurements of the average deposition width $w$ and the inclination angle of deposited layer $\theta$ were taken. The average deposition width $w$ for the T15 molten pool is narrower at 3.81 mm, compared to 4.29 mm for the T15 + Ce molten pool; the inclination angle $\theta$ of the T15 deposition layer is 140°, smaller than the 151° inclination angle $\theta$ of the T15 + Ce deposition layer, which aligns with observations from single-pass depositions.

Additionally, the average deposition height $h$ of the T15 molten pool was 1.52 mm, which was higher than the 1.17 mm average height observed for the T15 + Ce molten pool. The inclination angle between deposited passes $\omega$ in the T15 molten pool was 40°, which is lower than the 54° inclination angle between deposited passes observed in the T15 + Ce molten pool. Observations from Fig. 16(a1, b1) indicate that on the left side of the fusion line area, points P1 and P2 reached maximum velocities of $0.108 m/s$ and $0.0246 m/s$, respectively, with the velocity at P1 significantly exceeding P2. This suggests that the left fusion line area of the T15 molten pool exhibited a stronger velocity field, enhancing the climbing capability compared to the T15 + Ce molten pool, thereby resulting a greater deposition height and a steeper fusion line. Furthermore, the T15 molten pool exhibited a single inward flow with a deeper high-temperature zone and a steeper substrate fusion line. In contrast, the right side (R1 region) of the T15 + Ce molten pool was

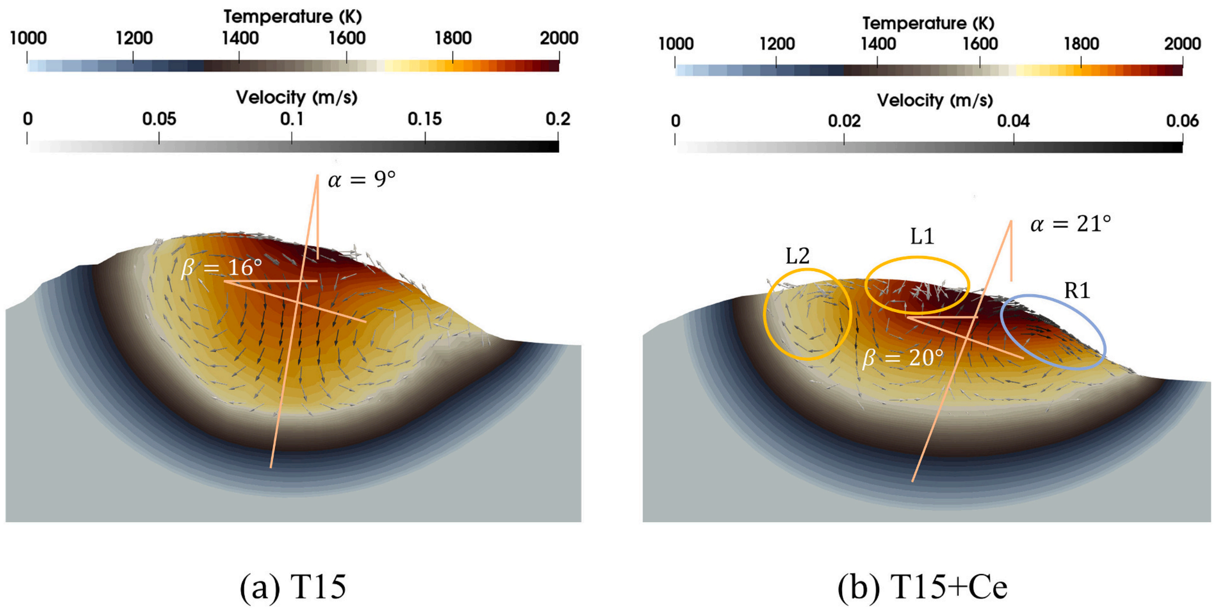



**Fig. 15.** Cross section distribution of flow and temperature in multi-pass deposition layer.

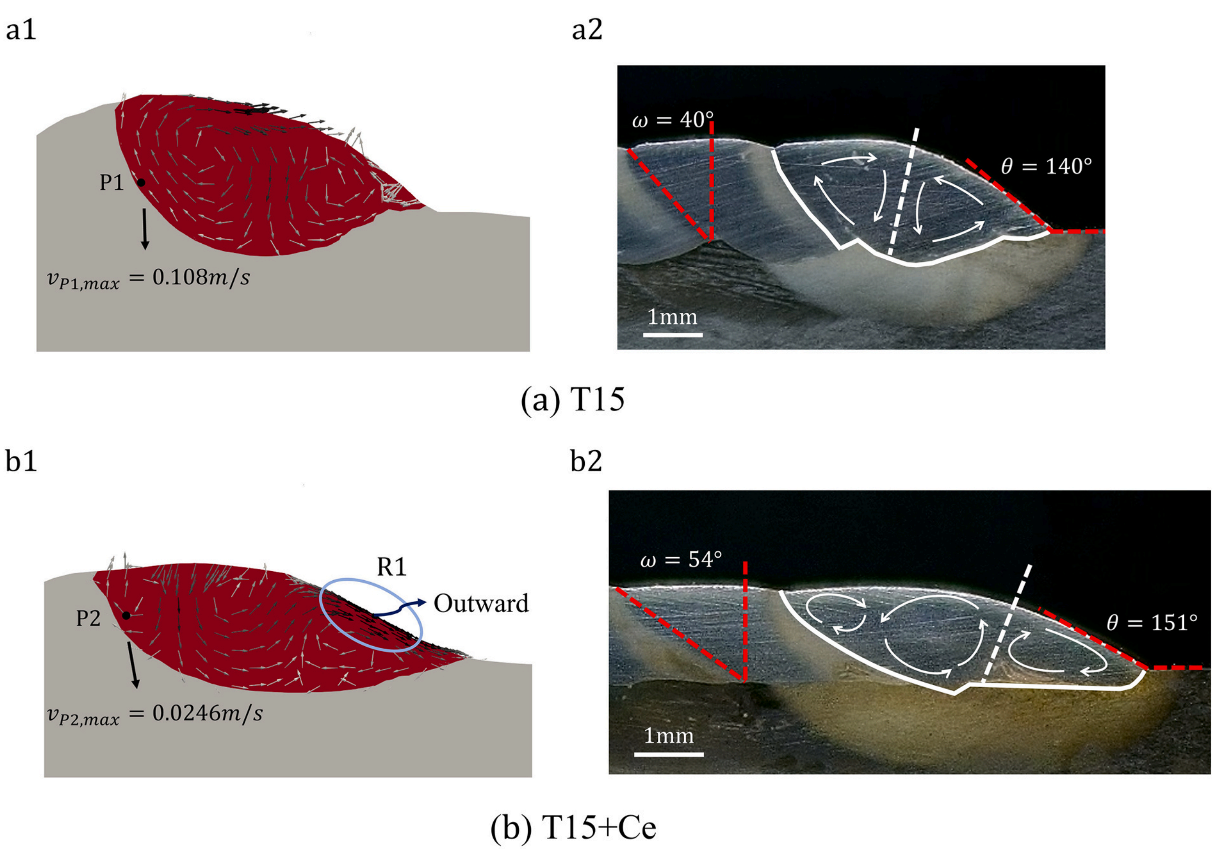



**Fig. 16.** Comparison of T15 and T15 + Ce molten pool morphology with experiments.

primarily influenced by outward flow, leading to a shallower high-temperature zone and a flatter substrate fusion line.

## 5. Conclusion

This study introduces a three-dimensional multiphase flow model based on the lattice Boltzmann method to simulate the evolution of molten pools in the laser-directed energy deposition process. The effectiveness of the model has been validated through comparisons with results from high-speed camera experiments and metallographic analyses of the deposition experiments. The following conclusions are drawn:

(1) The model effectively addressed numerous critical physical processes in the laser metal deposition process. An SRT-LB model with a mass source term has been used for the velocity field, and an enthalpy-based TRT-LB model with an energy source term has been developed to obtain the temperature field. The FSLBM was used to capture the gas-liquid free surface and addressed the Marangoni effect due to temperature gradients at interfaces. Moreover, the partially saturated method was utilized to revise the solid-liquid phase interface, ensuring the implementation of a no-slip condition at this interface.
(2) Quantitative comparisons between the simulation results of laser spot welding and high-speed camera results demonstrate consistency in key parameters such as the size of flow vortices, the ratio of inward to outward flow vortices, maximum velocity locations, and molten pool size. The simulation results indicated that the addition of Ce to the T15 powder altered the molten pool flow pattern, transitioning from a singular inward flow to a double-loop circulation pattern. At the transition temperature ($T_x$), a local stagnation in the velocity field was observed. Additionally, the interaction between inward and outward flows reduced the tangential velocity of the liquid metal, leading to a significantly lower peak flow velocity in the T15 + Ce molten pool compared to the T15 molten pool.
(3) In the single-pass laser deposition simulation, the aspect ratio ($h/w$) of the morphology parameters aligned with results from metallographic experiments, exhibiting errors of 5.0% and 5.9%, respectively. The flow pattern of the molten pool during the single-pass laser deposition process was identified as a primary factor influencing the morphology of the deposited layer. The formation of a high-temperature outward flow area in the T15 + Ce molten pool facilitated the redistribution of heat within the pool, pushing most of the heat outward to heat a larger area of the substrate. This led to a significant increase in deposition width ($w$), while the reduced tangential velocity decreased the liquid metal's climbing capability, resulting in a lower deposition height ($h$) and the formation of a wide, flat molten pool.
(4) During the multi-pass laser deposition process, asymmetry in the vortices was more pronounced in the T15 + Ce molten pool compared to the T15 molten pool. An inward flow region emerged at the left edge of the T15 + Ce molten pool, and the collision between inward and outward flows led to a significantly lower velocity field in the fusion line area of the T15 + Ce deposition layer compared to the T15 deposition layer. This resulted in a reduced deposition height and a flatter fusion line in the T15 + Ce deposition layer, consistent with metallographic results. Additionally, the right deposition area of the T15 + Ce molten pool was predominantly governed by outward flow, leading to a flatter substrate fusion line compared to the T15 deposition layer.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.icheatmasstransfer.2024.107859.

## CRediT authorship contribution statement

**Qihang Xue:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Data curation. **Gang Wang:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Yuelan Di:** Writing – review & editing, Funding acquisition, Conceptualization. **Lichao Liu:** Investigation. **Wei Shi:** Writing – review & editing. **Liping Wang:** Resources.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

## Acknowledgements

This work was supported by the National Natural Science Foundation of China (No. 52075282 & 52175207).

## Appendix A. The conversion factors between physical and lattice scale

The reference variables for physical scale are $\delta_x$, $\delta_t$, $\rho$, $T_{cold}$, $T_{hot}$. Between the physical scale and the lattice scale, these conversion factor can be calculated using the equations provided in Table A.1. The conversion factor for the variable $\phi$ is denoted as $C_\phi$. For most physical quantities, the conversion equation is as follows

$$\phi_{phys} = C_\phi{}^*\phi_{lattice} \tag{A.1}$$

To maintain a lattice scale ranging from 0.5 to 1.5, temperature conversion equation is as follows

$$T_{lattice} = \frac{\left(T_{phys} - T_{cold}\right)}{C_T} + 0.5 \tag{A.2}$$

**Table A.1**
The conversion factor between the physical scale and the lattice scale.

| Physical quantities | Physical units | Conversion factor |
|---|---|---|
| Length ($L$) | $m$ | $C_l = \delta_x$ |
| Time ($t$) | $s$ | $C_t = \delta_t$ |
| Density ($\rho$) | $kg/m^3$ | $C_\rho = \rho$ |
| Velocity ($u$) | $m/s$ | $C_u = C_l/C_t$ |
| Mass ($m$) | $kg$ | $C_m = \rho^* C_l{}^3$ |
| Kinematic viscosity ($\upsilon$) | $m^2/s$ | $C_\upsilon = C_l{}^2/C_t$ |

(continued on next page)

**Table A.1** (*continued*)

| Physical quantities | Physical units | Conversion factor |
|---|---|---|
| Force ($F$) | $kg \bullet m/s^2$ | $C_F = C_m{}^*C_l/C_t{}^2$ |
| Temperature ($T$) | $K$ | $C_T = T_{hot} - T_{cold}$ |
| Thermal diffusivity ($\alpha$) | $m^2/s$ | $C_\alpha = C_l{}^2/C_t$ |
| Specific heat capacity ($C_p$) | $J/(kg{\cdot}K)$ | $C_{C_p} = C_u{}^2/C_T$ |
| Thermal conductivity ($\lambda$) | $W/(m \bullet K)$ | $C_\lambda = C_F/(C_t{}^*C_T)$ |

## Appendix B. Supplementary materials

This includes eight videos in total: high-speed experiments for T15 and T15 + Ce; spot welding simulations for T15 and T15 + Ce; single-pass simulations for T15 and T15 + Ce; and multi-pass simulations for T15 and T15 + Ce.

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